



BEYOND BALANCING CALORIES

Companion Workbook



WEEK 1

UNDERSTANDING OBESITY



GOALS

- Understand the physiology of obesity, appetite, hormones, & weight set point
- Recognize why & how the body resists weight loss
- Learn how genes influence our susceptibility to obesity

TWO BRAINS IN ONE

HOW APPETITE & FOOD INTAKE ARE REGULATED

1. Cortico-Limbic System

Your "cognitive & emotional brain" absorbs info about your stress level & about the smell, taste, & texture of food. It causes you to "like" foods through opioid pathways & "want" foods through dopamine signals.

2. Hindbrain

Your "metabolic brain" senses & responds to the amount & quality of nutrients in your gut to determine when you're hungry or full.

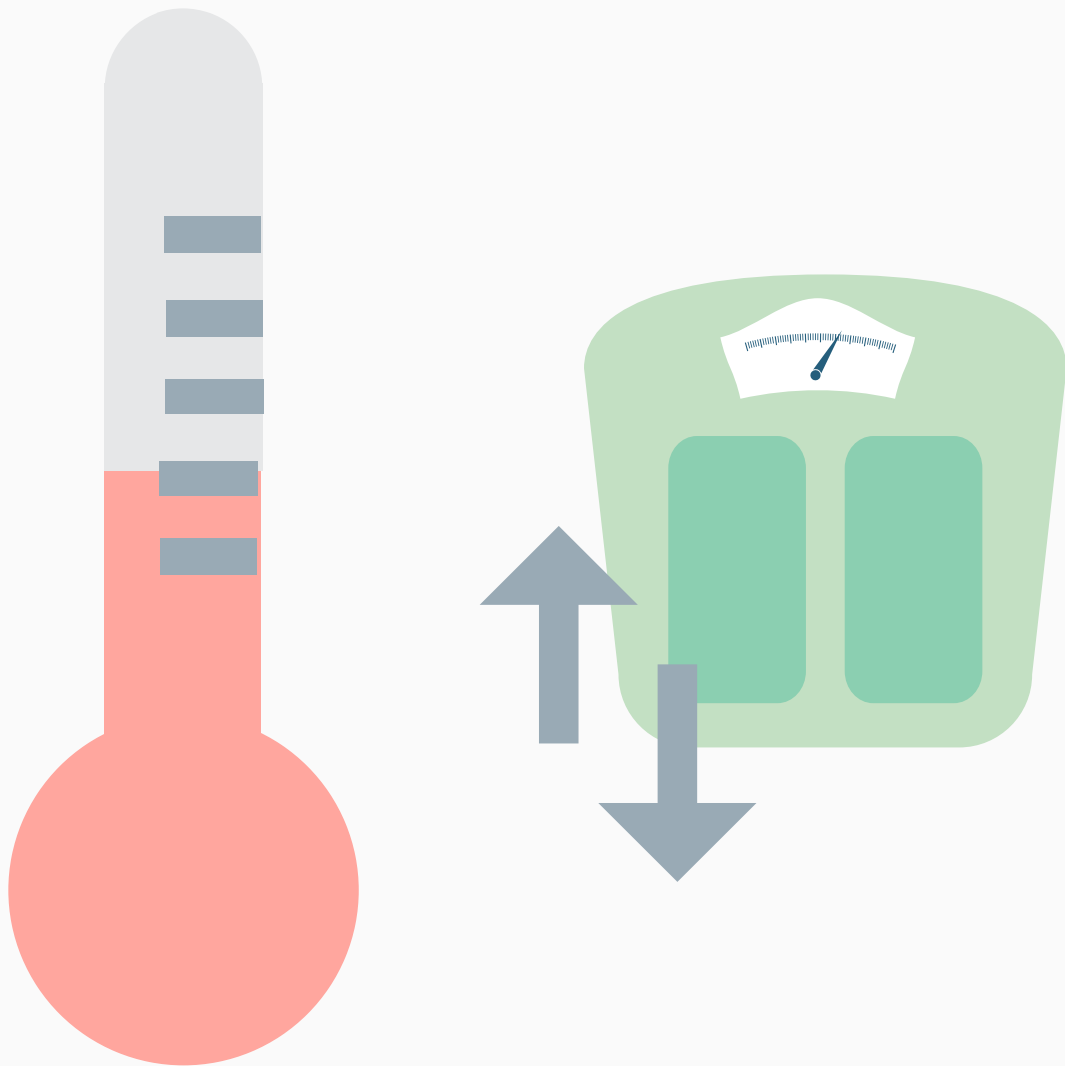
3. Hypothalamus

The hypothalamus combines info from both brains to regulate appetite & food intake. However, the cognitive & emotional brain can override the metabolic brain's signal that you're full.

WHAT IS SET POINT?

THE THERMOSTAT SETTING FOR YOUR WEIGHT

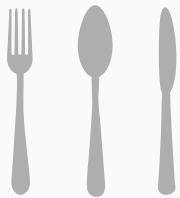
Your appetite & weight regulating system controls your calorie intake & metabolism to maintain your set point. It does this by sensing your food intake in the short term & fat stores, or adipose tissue, over the long term.



RESTRICTING CALORIES

THE EFFECTS IN OBESITY

Your set point is not fixed. It can get ratcheted up or down by environmental factors, such as your diet. Obesity is thought to be a disease of higher set point.



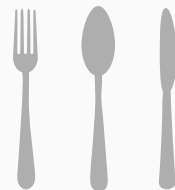
Calories Restriction

- Short-term weight loss
- Set point unchanged



Physiologic Compensation

- Appetite increases
- Metabolic rate decreases



Weight Regain

- Decreased dietary adherence
- Decreased ability to match reduced metabolic rate



HIGHER SET POINT

CIRCLE THE CORRECT ANSWER:

When you diet by cutting back on calories, you may lose weight in the short term. Yet, if you haven't worked at lowering your set point, your body will fight back. Your body will _____ **(increase, decrease)** the release of hunger hormones, such as ghrelin, and _____ **(increase, decrease)** satiety hormones (i.e., leptin and insulin), which will make you feel _____ **(more, less)** satiated but experience more pleasure and reward from food.

It will also _____ **(slow, speed up)** your metabolism. Over time, your powerful physiology will drive you to _____ **(lose, regain)** the weight.



WEEK 2

MANAGING OBESITY: THE ROLE OF DIET



GOALS

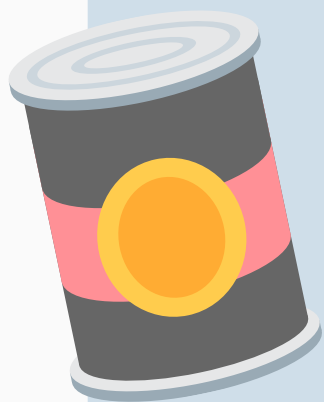
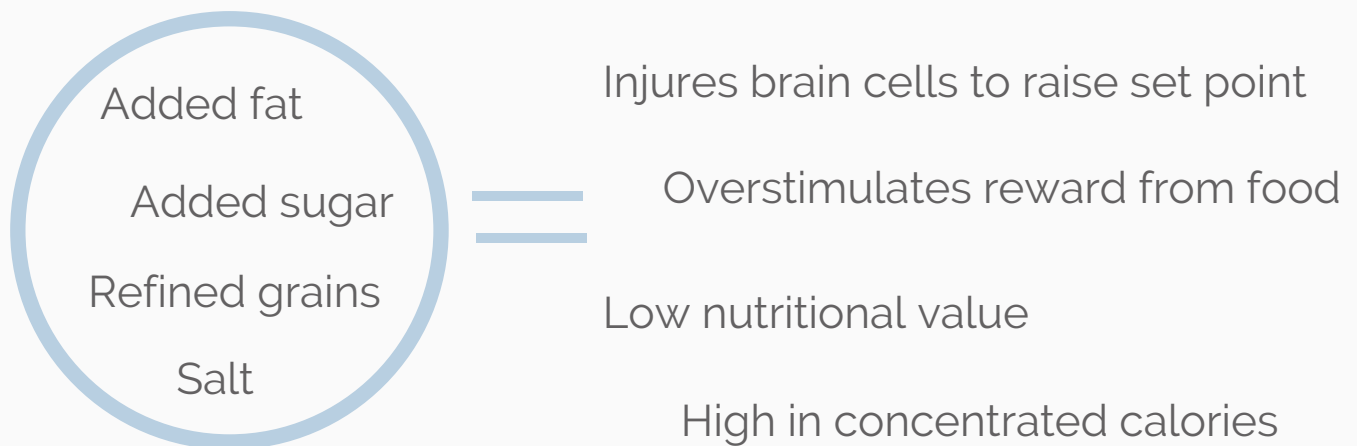
- Learn which foods overwhelm our reward system & raise set point
- Differentiate between unhealthy processed foods & whole foods
- Differentiate high vs low quality carbs, fats, & protein
- Find ways to increase whole foods in your diet long term

PROCESSED FOODS

WHAT ARE THEY AND HOW DO THEY AFFECT US?

Processed food is any food that has been altered from its natural form. While not all processed foods are unhealthy, ultra-processed foods have been changed the most from their natural form & are the least healthy.

What makes processed food unhealthy?



Minimally Processed

Canned Beans

Bagged Salad

Roasted Nuts

Frozen Vegetables

Ultra Processed

Desserts & Chips

Frozen Entrees

Sugary Beverages

Fast Food & Deli Meats



PROCESSED FOOD & APPETITE

CIRCLE THE CORRECT ANSWER:

We eat until stretch receptors in our stomach _____ **(reach a particular volume of food, until we have eaten a set amount of calories)**. Processed foods are _____ **(high, low)** in fiber, making them _____ **(high, low)** in volume. Thus, we consume more calories until we feel full.

Nutrients, such as vitamins and minerals _____ **(do, do not)** affect appetite. The lower nutrient density of processed foods causes us to overeat before we're full.

Saturated fats & sugars _____ **(increase, decrease)** the reward we get from food & _____ **(raise, lower)** set point. As a result, processed foods high in added fat, grains, sugar, & salt interact with our body to resist weight loss.

WHOLE FOODS

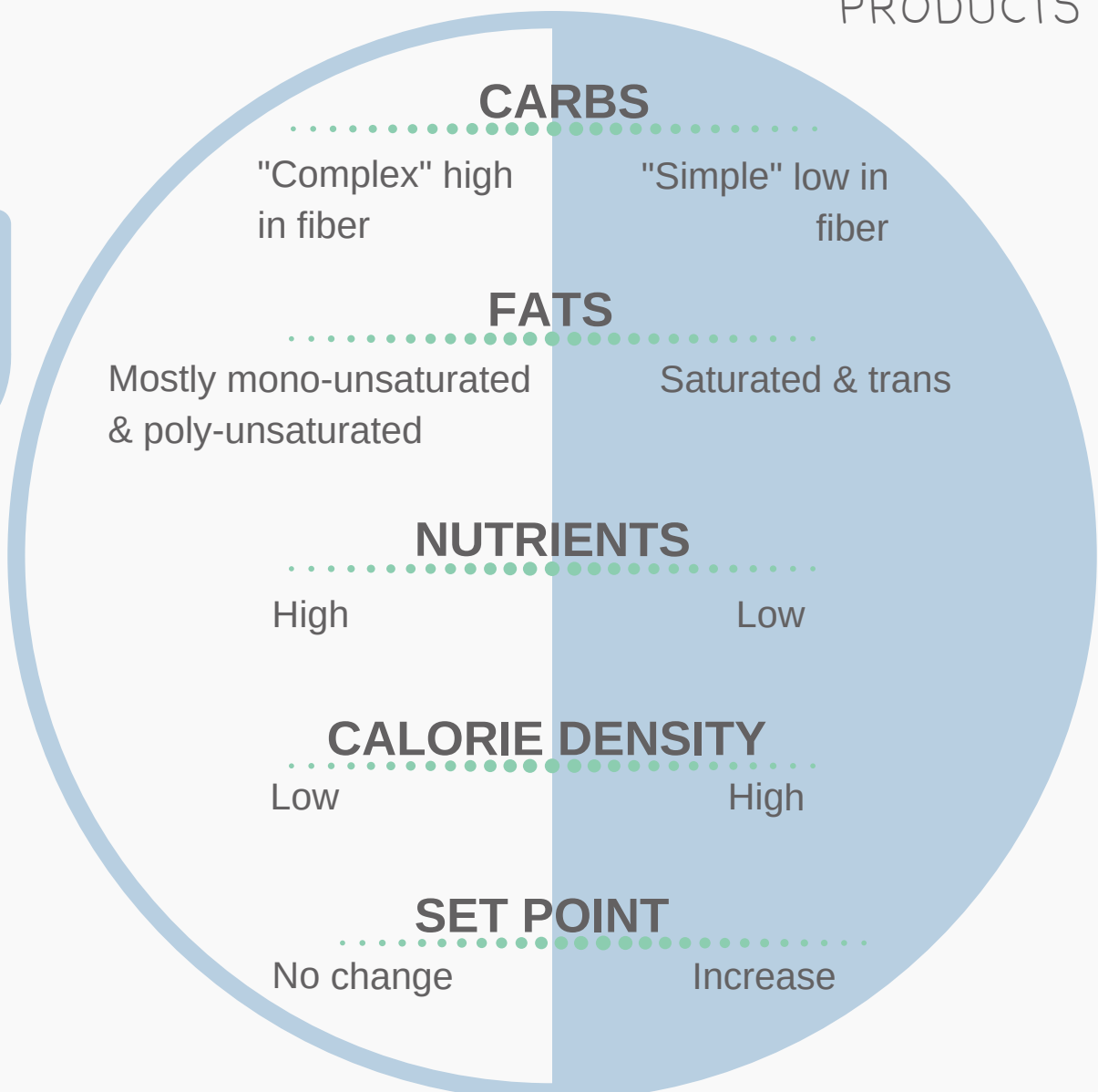
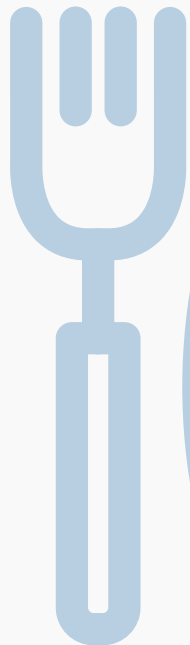
WHAT THEY ARE & HOW THEY DIFFER FROM PROCESSED FOODS

Whole foods are foods closest to their natural form:
fruits, vegetables, beans, nuts, & seeds

WHOLE/
PLANT-BASED

VS.

PROCESSED
& MOST ANIMAL
PRODUCTS



GETTING TO GREEN ON A WHOLE FOOD DIETARY PATTERN



Leafy & Non-Starchy Veggies

Spinach, Kale, Broccoli,
Cauliflower, Zucchini, Pepper

Legumes & Beans

Black Bean, Chickpea, Lentil,
Kidney Bean, Green Pea,
Soybean

Whole Grains

Quinoa, Millet,
Buckwheat, Brown Rice,
Oat, Barley

Processed & Regular Fat Animal Products

Hamburger, French Fries,
Chips, Mac & Cheese,
Pop Tart, Pizza

Fruit

Apple, Blueberry, Strawberry,
Mango, Pineapple, Pear

Starchy Veggies

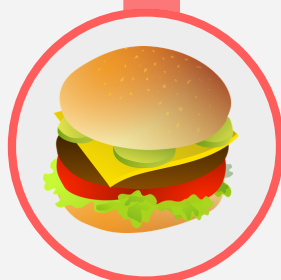
Yam, Acorn Squash, Pumpkin,
Corn, Potato, Parsnip

Nuts & Seeds

Walnut, Almond, Pecan,
Pumpkin Seed, Chia, Flax

Lean Animal Products

Non-Fat Yogurt, Egg White, Fish,
Non-Fat Milk, Lean Turkey,
Skinless Poultry



SELF-ASSESSMENT

How Much Processed vs. Whole Food Is In Your Diet?

In the last day, what did you have for:

BREAKFAST

LUNCH

DINNER

SNACKS

Which of the foods you listed were processed vs. whole?

PROCESSED

WHOLE

DESIGNING YOUR BEST WEIGHT LOSS DIET

Principle 1

The quality vs the quantity of fat, carbohydrate, & protein matters more for weight loss and health.

Carbohydrates: High Quality	Carbohydrates: Low Quality
• High in fiber, slowly digested • Minimally processed • Slow, small effect on blood sugar • Ex. Whole grains, fruits, veggies, legumes	• Low in fiber, rapidly digested • Highly processed • Fast, big effect on blood sugar • Ex. White breads, cereal, pasta, cakes, cookies, sugary drinks
Fats: High Quality	Fats: Low Quality
• Unsaturated fat (poly- & mono-unsaturated fat) • Ex. Nuts, seeds, olives, avocado, fish	• Saturated & trans fats • Ex. Meats, processed meats, fast foods, full-fat dairy, cheese, butter
Protein: High Quality	Protein: Low Quality
• Low in saturated fat & high in nutrients • Ex. Beans, lentils, lean fish & poultry	• High in saturated fat & low in nutrients • Ex. Fatty meats, full-fat dairy, cheese

DESIGNING YOUR BEST WEIGHT LOSS DIET

Principle 2

The best weight loss diet for you is one that you can stick with long term - one that matches your preferences, traditions, & customs.

Sample Meal Plan

	Day 1	Day 2	Day 3
Breakfast	Overnight rolled oats	Sprouted grain bread w/ p.b.	Fresh fruit w/ non-fat Greek yogurt
Lunch	Mixed green salad w/ split pea soup	Curried lentils	Stuffed pepper + wild rice & beans
Dinner	Teriyaki tofu w/ steamed vegetables	Salmon w/ pesto	Vegetarian chili
Snacks	Sliced apple	Hummus w/ vegetables	Handful of almonds



TRANSITIONING TO YOUR BEST WEIGHT LOSS DIET

Think of your cultural preferences, traditions, food allergies, & health conditions. From mostly plant-based whole foods, what foods could you substitute or add to your diet?

Week #2 Challenge: Create Your Own Whole Food Meal Plan

	Breakfast	Lunch	Dinner	Snacks
Monday				
Tuesday				
Wednesday				
Thursday				
Friday				
Saturday				
Sunday				



WEEK 3

BEYOND DIET: THE ROLE OF STRESS, EXERCISE, & SLEEP



GOALS

- Understand how stress, sedentary lifestyle, & inadequate sleep interfere with weight loss
- Learn stress management techniques that improve your response to stress
- Assess whether you're getting the recommended amount of physical activity
- Identify factors interfering with getting the recommended amount & quality of sleep
- Set a goal for overcoming your biggest hindrance to weight loss

STRESS & WEIGHT MANAGEMENT

- Stress drives us to eat comfort food to soothe negative feelings & chemically dial down our stress response.
- Stress hormones lead to an accumulation of belly fat - which releases chemicals that cause inflammation & insulin resistance, leading to further weight.
- Stress triggers can be EXTERNAL (related to a job, relationships, or major life changes) or INTERNAL (challenges to health or mental wellbeing).
- Stress is part of our lives. We can't always change the situation that causes stress. By recognizing triggers, you can become better at responding to stress.
- Managing stress is an integral part of controlling weight.

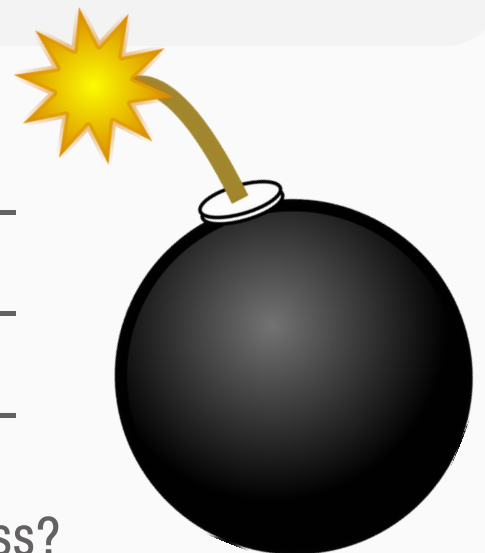
List 3 of your stress triggers

1. _____
2. _____
3. _____

How would you rate how you respond to stress?

1 2 3 4 5 6 7 8 9 10

I feel little stress I always feel stressed



EXERCISES FOR BUILDING STRESS RESILIENCE

- Managed stress can be healthy & help you overcome obstacles.
- Feeling high levels of stress, however, can be harmful to your weight & your health.
- Stress management techniques that build resilience to stress can lower our level of stress hormones & improve how we regulate our appetite & weight.



Build Optimism



Express Gratitude



Invest in Relationships



Find Meaning & Purpose

LEARNING OPTIMISM

"Seeing the Cup Half Full"

Negative thoughts are a barrier to achieving any goal. Transforming negative self-talk into positive affirmations can empower & give you confidence to overcome challenges. By paying attention to & transforming your response to adversity, you can perceive stress positively.

1. Recall a negative thought you had in a recent stressful situation (e.g., feeling you weren't good enough)
2. Challenge that thought (e.g., Why do you not feel confident? In which ways *are* you qualified?)
3. Substitute a positive thought (e.g., I have the skills to do this, I will learn a lot)
4. Do this once a day each day this week

Negative Thoughts

Positive Thoughts



JOURNALING GRATITUDE

Cultivating gratitude is the practice of focusing on what you have rather than what is missing in your life. The weekly habit of writing what you are grateful for in a journal is another way of reducing your stress level.

List 2 things for which you are thankful this week (they can be a person, event, or a pleasure.) Try to be detailed.

POWER YOUR METABOLISM THROUGH EXERCISE

Your weekly snapshot

How many days per week do you do aerobic exercise?
(walking, jogging, dancing, etc.) _____

How many minutes do you average a session? _____

Calculate your minutes per week of aerobic exercise
(days per week x minutes each time) _____

How many days per week do you do strength
resistance exercise? (yoga, weights, body
resistance, etc.) _____

Are you getting the recommended amount of
exercise?

Do you perform ≥ 150 minutes/week of
aerobic exercise? YES NO

Do you do strength resistance exercises
 $\geq 2x$ /week? YES NO

What are ways you can incorporate more activity into
your day? _____



SLEEP INVENTORY

On average, how many hours of sleep do you get each night? _____

Are you getting 7-9 hours of sleep per night? YES NO

How would you rate the quality of your sleep?
POOR FAIR GOOD EXCELLENT



Check all factors that may be interfering with your sleep goals

Conditioning

- Do not have a consistent bed & wake time
- Use bed other than for sleep & sex
- Stay in bed if can't sleep > 20 minutes

Habits

- Consume late day caffeine & excess alcohol
- Lack of exercise
- Use of elec. devices within 30 min. of bedtime

Sleep Environment

- Bedroom is not quiet, dark, & cool
- TV & pets in bedroom
- Cell phone in bedroom

Relaxation Techniques

- Do NOT use any of the following:
- Meditation or relaxing music
- Guided imagery

CREATING A PERSONAL GOAL

Week #3 Challenge: Create a Personal Goal to Overcome Your Biggest Hindrance to Weight Loss

S	M	A	R	T
Specific	Measurable	Actionable	Realistic	Time-based
Write down a specific goal (detailed is best)	When will you work on your goal & track your progress?	How will you achieve your goal?	Do you have time & resources to meet your goal?	When will you start? Do you have a deadline(s)?



WEEK 4

USING HABIT & EMOTIONS TO YOUR ADVANTAGE



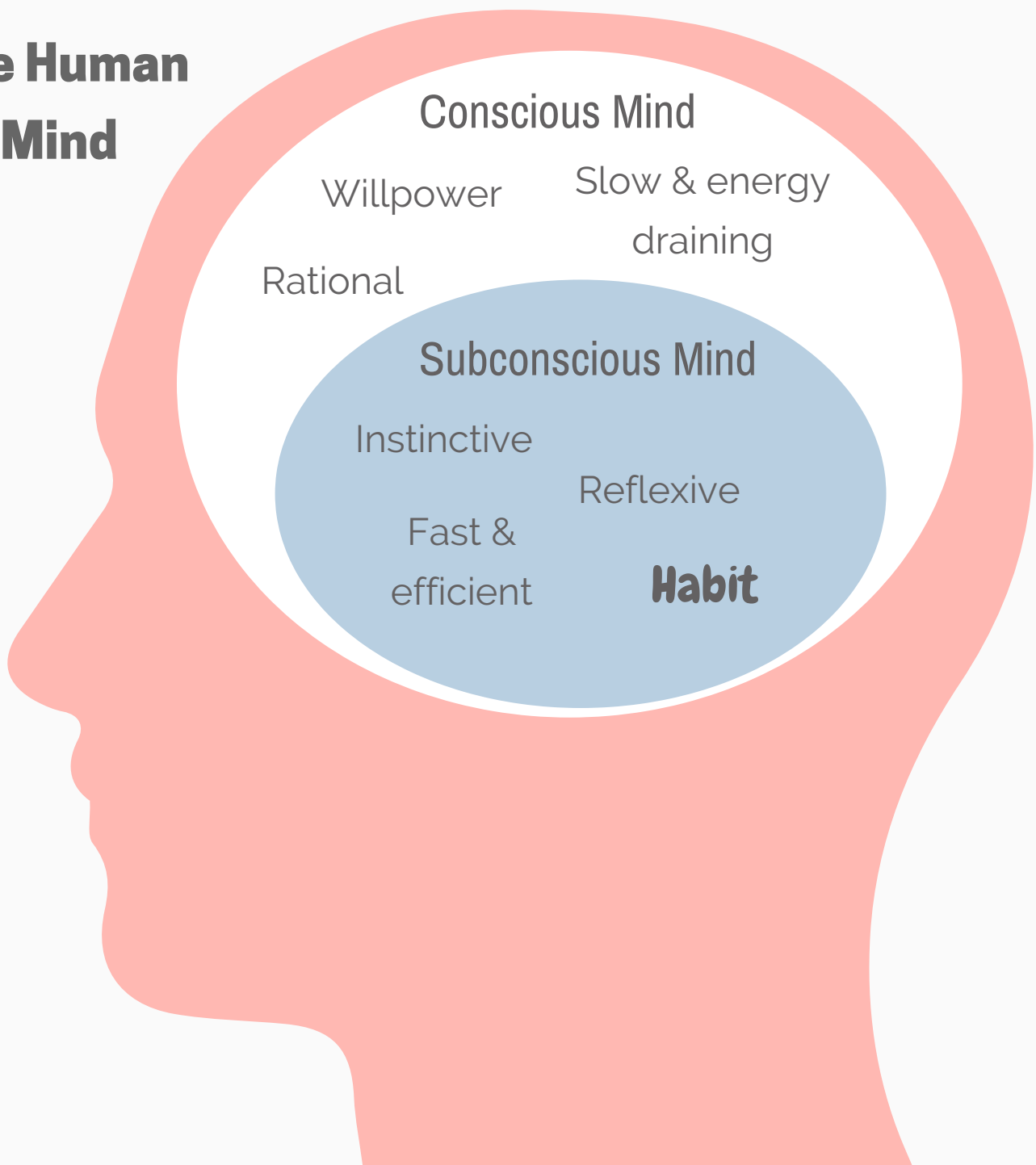
GOALS

- Recognize that many choices are made subconsciously
- Associate a daily cue with a health behavior you want to adopt
- Intentionally design your surroundings to make healthy choices the easiest & most convenient
- Practice mindfulness when you eat

THE HUMAN MIND

- Knowledge about healthy habits cater to our conscious mind.
- Most of our food choices are made instinctively by our subconscious mind.
- We need to use our conscious brain to structure our habits & environment so that we make healthy subconscious choices.

The Human Mind



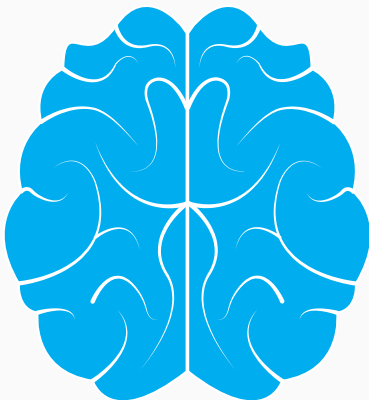
MIND OVER MATTER

The main way our subconscious mind makes choices are through:

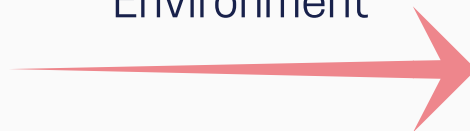
1. Habits, which are choices triggered by a cue in anticipation of a reward.
2. Default choices, based on the path of least resistance.

Both can be used intentionally to our advantage.

Subconscious Decision



Designing Your Environment



Habit



Healthy Choice



What are some of your current habits?

Try to become aware of the context that prompts these habits

CREATING NEW HABITS



- Become aware and disrupt the association between the cue & a habit you want to break this week
- Create an association with something you do every day & the new habit you want to build
- Lather, rinse, repeat to reinforce this new association daily

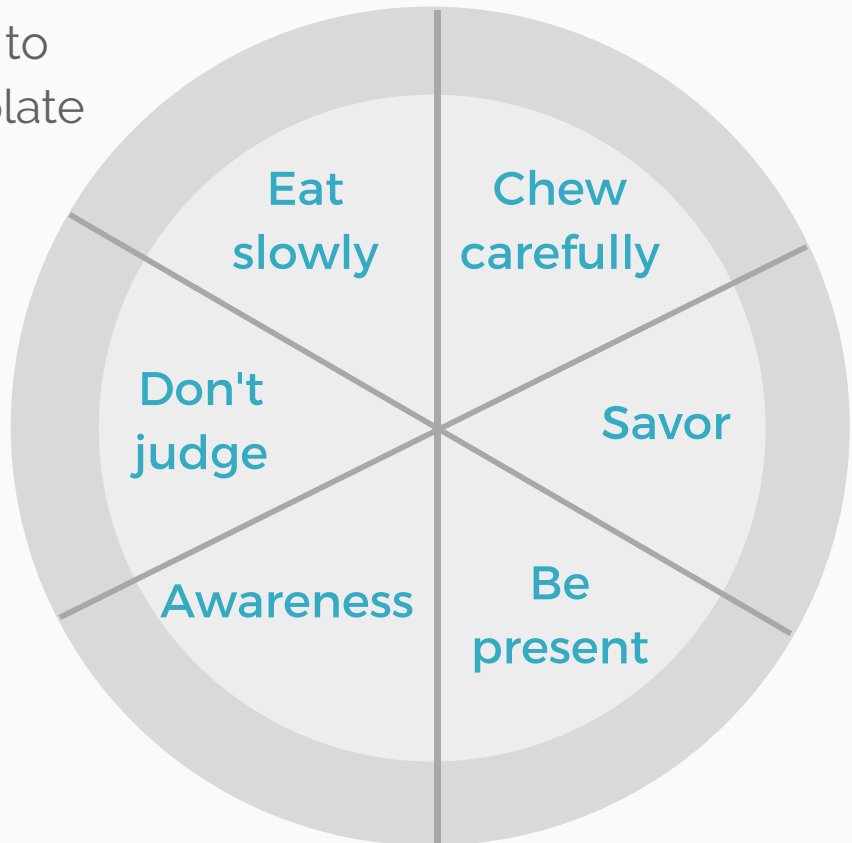
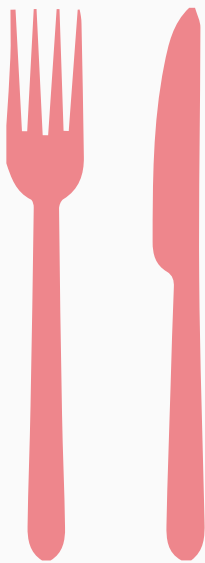
List a new habit you would like to form:

List a way you plan to make a healthy choice the one that is most convenient this week (e.g., have handful of nuts readily available when you are tired, hungry, or stressed.)

EAT MINDFULLY

Week #4 Challenge: Creating a Mindful Plate

Follow these tips to
fill your mindful plate



Check each day this week you were mindful during meals:

☐ S ☐ M ☐ T ☐ W ☐ TH ☐ F ☐ S

What difference have you noticed? Do you feel full faster? Do you feel satisfied after meals? Do you taste food differently?

WEEK 5
CREATING A
WEIGHT-LOSS PLAN: A
STEP-WISE APPROACH



GOALS

- Set a realistic weight-loss goal
- Follow steps to unlock your set point
- Design your environment to support your vision & goal

SET A WEIGHT-LOSS GOAL

Complete the following three steps to determine your appropriate weight-loss amount:

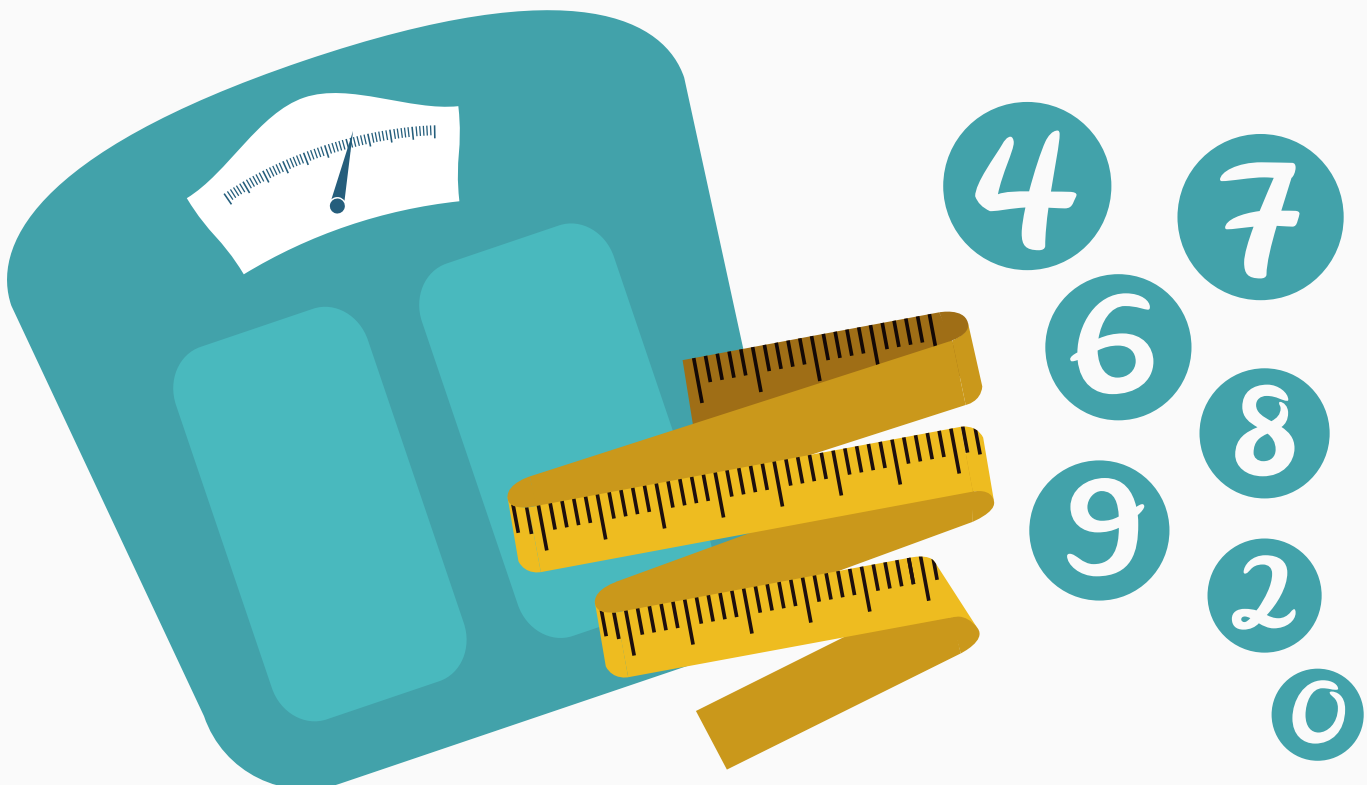
1. What is your current weight? _____ lbs

2. Calculate 5% of your weight:

.05 x _____ (your weight) = _____

3. Subtract the amount in step 2 from your current weight to determine your target amount of weight loss:

_____ (your weight) - _____ (5% of your weight) = _____ (weight-loss goal)



YOUR WEIGHT LOSS PLAN

1. Change your dietary pattern by choosing high-quality sources of macronutrients.*

Swap refined processed carbohydrates (bread, pasta, muffins, desserts) with fiber-rich sources (fruit, vegetables, whole grains)

List 2 changes you can make:

Swap animal sources of fat (meat, chicken, cheese, butter) with plant sources (avocados, nuts, seeds)

List 2 changes you can make:

2. Change behaviors that may affect your set point, such as sleep, exercise, or stress. You'll know which one unlocks your set point when your weight comes off & stays off with less effort.**

What are areas in which you can improve?

* After changing quality, experiment with each macronutrient's quantity

** If you aren't noticing weight loss, try combining changes in your diet & behaviors. This may overcome overlapping ways your body compensates in response to weight loss

SUPPORT YOUR SUCCESS

Follow these guidelines to create a microenvironment that helps you maintain your changes & achieve success,

1. Design your surroundings to support your goals

- Remove hurdles for the habits that align with your goals & personal vision
- Add steps for the ones you wish to change

2. Transform your personal food microenvironment, e.g., all the places where you eat your meals & snacks

3. Meal plan: plan & prep your meals for the whole week ahead of time

- Select recipes or restaurant meals (review menus for healthy options)
- Make a list and shop for groceries that you'll need
- Prep by chopping vegetables or marinating poultry or fish



FOOD ESSENTIALS

Here is a helpful list of some foods you should keep stocked in your microenvironment.

Pantry

- Sprouted whole-grain bread
- Intact whole grains (barley, oats, quinoa)
- Canned beans & tomatoes
- Spices (turmeric, cumin, garlic, oregano, ginger)
- Extra-virgin olive oil
- Vinegars
- Vegetable bouillon

Freezer

- Assortment of fruit
- Assortment of vegetables (medleys, steam bags, riced cauliflower)
- Veggie burgers
- Cooked whole grains
- Edamame

Fridge

- Non-dairy almond, soy, or rice milk
- Fresh fruit (washed & cut)
- Fresh vegetables (pre-washed, pre-cut)
- Hummus & salsa
- Non-fat plain yogurt
- Bagged, pre-washed spinach, kale, lettuce
- Tofu

Snacks

- Nuts (almonds, walnuts)
- Seeds (sunflower, pumpkin)
- Dried fruits (dates, apricots, cranberries, raisins)
- Peanut butter or other nut butters
- Plain popcorn

SELECT REFERENCES

Week 1

Fothergill E, Guo J, Howard L, et al. Persistent metabolic adaptation 6 years after The Biggest Loser competition. *Obesity* (Silver Spring, Md). 2016;24(8):1612-1619. doi:10.1002/oby.21538.

Kolata, G. After 'The Biggest Loser,' Their Bodies Fought to Regain Weight. *The New York Times*. May 2, 2016 (last accessed 1/07/2018)

Sumithran P, Prendergast LA, Delbridge E, et al. Long-Term Persistence of Hormonal Adaptations to Weight Loss. *New England Journal of Medicine*. 2011;365(17):1597-1604.

Guyenet SJ, Schwartz MW. Regulation of Food Intake, Energy Balance, and Body Fat Mass: Implications for the Pathogenesis and Treatment of Obesity. *The Journal of clinical endocrinology and metabolism*. 2012;97(3):745-55. doi: 10.1210/jc.2011-2525. PubMed PMID: PMC3319208.

Zheng H, Lenard N, Shin A, Berthoud H-R. Appetite control and energy balance regulation in the modern world: Reward-driven brain overrides repletion signals. *International journal of obesity* (2005). 2009;33(Suppl 2):S8-13. doi: 10.1038/ijo.2009.65. PubMed PMID: PMC2838178.

Berthoud HR, Munzberg H, Morrison CD. Blaming the brain for obesity: Integration of hedonic and homeostatic mechanisms. *Gastroenterology*. 2017. Epub 2017/02/14. doi: 10.1053/j.gastro.2016.12.050. PubMed PMID: 28192106.

SELECT REFERENCES

Week 1, cont.

Münzberg H, Qualls-Creekmore E, Yu S, Morrison CD, Berthoud H-R. Hedonics Act in Unison with the Homeostatic System to Unconsciously Control Body Weight. *Frontiers in Nutrition*. 2016;3:6. doi: 10.3389/fnut.2016.00006. PubMed PMID: PMC4753312.

Locke AE, Kahali B, Berndt SI, et al. Genetic studies of body mass index yield new insights for obesity biology. *Nature*. 2015;518(7538):197-206. doi:10.1038/nature14177.

Week 2

<https://www.ers.usda.gov/> (last accessed January 9, 2018)

Cordain L, Eaton SB, Sebastian A, Mann N, et al. Origins and evolution of the Western diet: health implications for the 21st century. *The American Journal of Clinical Nutrition*. 2005;81(2):341-54.

Thaler JP, Schwartz MW. Minireview: Inflammation and Obesity Pathogenesis: The Hypothalamus Heats Up. *Endocrinology*. 2010;151(9):4109-15. doi: 10.1210/en.2010-0336. PubMed PMID: PMC2940486.

<http://www.nytimes.com/2013/02/24/magazine/the-extraordinary-science-of-junk-food.html> (last accessed January 10, 2018)

Gross LS, Li L, Ford ES, Liu S. Increased consumption of refined carbohydrates and the epidemic of type 2 diabetes in the United States: an ecologic assessment. *Am J Clin Nutr* 2004 79: 5 774-779

SELECT REFERENCES

Week 2, cont.

<https://www.theatlantic.com/health/archive/2014/03/science-compared-every-diet-and-the-winner-is-real-food/284595/> (last accessed January 10, 2018)

Katz DL and Meller S. Can We Say What Diet Is Best for Health? Annual Review of Public Health 2014 35:1, 83-103

Sacks FM, Bray GA, Carey VJ, Smith SR, et al. Comparison of Weight-Loss Diets with Different Compositions of Fat, Protein, and Carbohydrates. N Engl J Med 2009;360:859-73.

Heianza Y, Ma W, Huang T, Wang T, et al. Macronutrient Intake–Associated FGF21 Genotype Modifies Effects of Weight-Loss Diets on 2-Year Changes of Central Adiposity and Body Composition: The POUNDS Lost Trial. Diabetes Care 2016;39:1909–1914

Week 3

Morris MJ, Beilharz JE, Maniam J, Reichelt AC, Westbrook RF. Why is obesity such a problem in the 21st century? The intersection of palatable food, cues and reward pathways, stress, and cognition. Neuroscience & Biobehavioral Reviews. 2015;58:36-45

Kuo LE, Kitlinska JB, Tilan JU, et al. Neuropeptide Y acts directly in the periphery on fat tissue and mediates stress-induced obesity and metabolic syndrome. Nature Medicine. 2007;13:803.

SELECT REFERENCES

Week 3, cont.

Wu G, Feder A, Cohen H, et al. Understanding resilience. *Frontiers in Behavioral Neuroscience*. 2013;7(10).

Dunstan DW, Howard B, Healy GN, Owen N. Too much sitting -- A health hazard. *Diabetes Research and Clinical Practice*. 97(3):368-376.

Spiegel K, Tasali E, Penev P, Cauter EV. Brief Communication: Sleep Curtailment in Healthy Young Men Is Associated with Decreased Leptin Levels, Elevated Ghrelin Levels, and Increased Hunger and Appetite. *Ann Intern Med*. 2004;141:846-850.

Week 4

Strack F, Deutsch R. Reflective and impulsive determinants of social behavior. *Personality and social psychology review: an official journal of the Society for Personality and Social Psychology, Inc*. 2004;8(3):220-247.

David TN, Wendy W, Jeffrey MQ. Habits—A Repeat Performance. *Current Directions in Psychological Science*. 2006;15(4):198-202.

Wood, W., Quinn, J. M., & Kashy, D. A. (2002). Habits in everyday life: Thought, emotion, and action. *Journal of Personality and Social Psychology*, 83(6), 1281-1297.

Münzberg H, Qualls-Creekmore E, Yu S, Morrison CD, Berthoud H-R. Hedonics Act in Unison with the Homeostatic System to Unconsciously Control Body Weight. *Frontiers in Nutrition*. 2016;3(6).

SELECT REFERENCES

Week 4, cont.

Marteau TM, Hollands GJ, Fletcher PC. Changing human behavior to prevent disease: the importance of targeting automatic processes. *Science* (New York, N.Y.). 2012;337(6101):1492-1495.

O'Reilly GA, Cook L, Spruijt-Metz D, Black DS. Mindfulness-Based Interventions for Obesity-Related Eating Behaviors: A Literature Review. *Obesity reviews: an official journal of the International Association for the Study of Obesity*. 2014;15(6):453-461.

Week 5

Fildes A, Charlton J, Rudisill C, Littlejohns P, Prevost AT, Gulliford MC. Probability of an Obese Person Attaining Normal Body Weight: Cohort Study Using Electronic Health Records. *American Journal of Public Health*. 2015;105(9):e54-e59.

Bucher T, Collins C, Rollo ME, et al. Nudging consumers towards healthier choices: a systematic review of positional influences on food choice. *The British journal of nutrition*. 2016;115(12):2252-2263

Larson N, Story M. A review of environmental influences on food choices. *Ann Behav Med*. 2009;38 Suppl 1:S56-73.